MOHSIN ALI

Chief Technology Officer | AI, ML & Quantum Systems | Deep-Tech R&D & Engineering Leadership

Dhaka, Bangladesh | alimohsin.dev0@gmail.com | linkedin.com/in/imalimohsin | alimohsinofficial.github.io

EXECUTIVE SUMMARY

Technology executive and hands-on engineering leader with 10+ years building software, AI/ML, and deep-tech products from research to production. As Chief Technology Officer at Quantonova, I lead a portfolio of Innovate UK–funded innovation programmes spanning AI-driven industrial monitoring, quantum sensing, edge inference, and agentic automation. Proven record of architecting production-grade systems, winning competitive R&D funding, publishing peer-reviewed research, and translating emerging technologies — from quantum error correction to physics-informed machine learning — into commercially viable, standards-compliant products. Equally fluent across the full stack and at the executive table: strategy, delivery, team-building, and cross-functional partnership.

CORE COMPETENCIES

Technology Strategy & Roadmaps • AI / ML & Agentic Architecture • Innovate UK R&D Leadership • Full-Stack & Cloud Engineering • Quantum Sensing & Error Correction • Edge AI & Embedded Inference • Data Engineering & Analytics • Product Commercialisation • Standards & Regulatory Compliance • Team Leadership & Mentoring

PROFESSIONAL EXPERIENCE

Chief Technology Officer & Project Manager — 2023 – Present

Quantonova Ltd · *Innovate UK R&D portfolio (AI, quantum sensing, advanced manufacturing)*

- Lead technology strategy and delivery across a multi-project R&D portfolio, partnering with academic and industry consortia (Nottingham Trent University, MHK Consulting, University of Waterloo, Ocean Sonics).
- **QuantAirSense — AI compressed-air monitoring:** architected an ML platform detecting compressed-air leaks, equipment degradation, and scheduling waste for UK SME manufacturers — Random Forest classifier (F1 = 0.99, 1% false-positive, zero false-negative) with SHAP explainability and forecasting over 15 engineered temporal features.
- Quantified ~£1,900/yr recoverable savings and ~2 t/yr CO_2 reduction per site; validated against ISO 11011:2013, Carbon Trust CTV044, DOE CAC and IEC 60300-3-11; deployed a SCADA-style monitoring dashboard on Google Cloud Run (europe-west2).
- **AI-AFS — acoustic AI for aquaculture:** led algorithm development for an underwater-sound system detecting salmon feeding behaviour in recirculating-aquaculture systems; built classifiers on 878 real hydrophone clips achieving 0.99 accuracy and AUC 0.999, owning the core work-package deliverables.
- Developed physics-informed neural-network models for bioprocess yield optimisation (R^2 = 0.913, +29.7% optimisation gain) and federated graph-learning concepts for supply-chain resilience.
- Authored and won competitive Innovate UK grant applications; produced FPGA / edge-inference feasibility studies (PYNQ-Z2) to move trained ML models towards production deployment.

– Built an internal 48-module project-management platform (Laravel, Livewire, Tailwind CSS) to coordinate delivery, documentation, and compliance across the portfolio.

Chief Technology Officer & Software Engineer 2020 – 2023
143Play · *Game & software studio*

– Set and executed the company's technical vision and architecture across game and software products; led engineering direction, code standards, and toolchains.
– Drove product delivery from prototype to release, balancing long-term technical strategy with hands-on engineering ownership.

Software Engineer & Game Developer 2016 – 2020
143Play

– Designed and shipped games and software across web, Android, and desktop, including published Android titles featuring Bluetooth multiplayer and real-time gameplay.

Game & Web Developer 2014 – 2016
143Play

– Built web applications and game prototypes; co-developed a lightweight JavaScript-based 3D game-rendering engine.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, PHP, Java, C / C++, SQL
AI / ML & Data Science	Exploratory Data Analysis (EDA), Random Forest, SHAP explainability, Physics-Informed Neural Networks (PINN), edge & embedded inference, prompt engineering, scikit-learn, pandas / NumPy
AI Coding Agents & Tools	Claude Code, OpenAI Codex, Google Gemini, NotebookLM, n8n (agentic / workflow automation)
Frontend	React, React Native, Vue, Web Components, Tailwind CSS, HTML5 / CSS3, Figma
Backend, Cloud & DevOps	Node.js, Laravel, Livewire, CodeIgniter, Flutter, Docker, Google Cloud Platform (Cloud Run), REST APIs, CI/CD, Git
Databases	PostgreSQL, MySQL, SQLite
WordPress & CMS	WordPress theme & plugin development, Bricks Builder, Elementor
Data & BI	Power BI, Jupyter, data visualisation & SCADA-style dashboards
Quantum & Edge Hardware	Quantum Error Correction, NV-centre quantum magnetometry, quantum LiDAR sensing, FPGA edge inference (PYNQ-Z2)
Design & Media	Adobe Photoshop, Adobe Illustrator, DaVinci Resolve, Adobe Premiere Pro
Game Development	Unity, Unreal Engine, Blender, World Creator, Gaea
Productivity & Tooling	Obsidian, Notion, Chrome Extension development

EDUCATION

M.Sc. in Computer Science & Engineering 2017
Jahangirnagar University, Savar, Bangladesh

B.Sc. in Computer Science & Engineering 2016
Daffodil International University, Dhaka, Bangladesh

SELECTED PUBLICATIONS & RESEARCH

- **QuantoTrace: Quantum Error Correction as a Service for Robust Quantum Computing** — Co-author (3rd). IEEE Xplore
- **A Novel Method to Predict Water Quality Resilience using Deep Reinforcement Learning in São Paulo, Brazil** — Co-author (2nd). IEEE Xplore
- ICCIT 2014 Student Conference (3 papers): UX-driven design methods; logo & GUI design psychology for applications; lightweight JavaScript 3D game-rendering engine.

ACHIEVEMENTS & ACTIVITIES

- ACM ICPC Asia Regional (Dhaka Site) — 2014 (Rank 71) and 2015 (Rank 84), Team DIU_PEKKA.
- Published Android developer portfolio (2016) — multiple released titles spanning card games, puzzles, and utility apps.

LANGUAGES

English — Reading: Advanced | Writing: Intermediate | Speaking: Intermediate

REFERENCES

Prof. Dr. Syed Akhter Hossain — Dean, Faculty of Science & IT, Daffodil International University, Bangladesh aktarhossain@daffodilvarsity.edu.bd | linkedin.com/in/dr-syed-akhter-hossain

Narayan Ranjan Chakraborty — PhD Research Fellow, University of Agder (UiA); Associate Professor & Associate Head, Daffodil International University narayan@daffodilvarsity.edu.bd | linkedin.com/in/narayan633

Dr. Mahmudul Hasan — COO & Lead Data Scientist, MediprospectsAI; Sr. Faculty, Anglia Ruskin University, UK mahmudul.05@gmail.com | linkedin.com/in/mmhplus